ICS 604
Applied Data Science

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Reminder: Project proposal

- Due: **11:59 PM, today (Monday, March 23)**
  - Create a **PDF file** and submit it to Lamaku.
  - Maximum 2 pages (~ 700 – 900 words)
- The proposal **must include a title.**
- You may include **as many figures as needed**; they do not count toward the page limit.
- The proposal will be evaluated based on **how well it addresses the items in the project requirements document**, not on the idea or scope itself.

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Project proposal review

- Two proposals will be assigned tomorrow.
- Peer feedback is due by **11:59 PM on Monday, March 30.**
  - Write approximately half a page to one page per proposal.
    - Include the title of the project you reviewed.
  - Create a **separate PDF file for each proposal reviewed** and submit it to Lamaku under "Proposal Peer Feedback" in Assignments tool.
    - Name each file using the format: "[YourLastName]_feedback_[Proposal_ID].pdf"
    - Do NOT include your name in the file.
  - Refer to the project requirements document for guidance.
  - Your feedback will be anonymized and shared with the proposal author.

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Lecture 16

- Hypothesis Testing
  - Comparing means (cont'd.)
  - Comparing proportions
  - Comparing multiple categories

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Previously… Hypothesis testing

- A statistical technique for making inferences about populations using sample data, often by comparing two or more samples to determine whether observed differences are statistically significant.
- Two main hypothesis:
  - **Null hypothesis (H_0):** Assumes no change, difference, or effect — any observed result is due to random chance. Serves as the baseline for testing.
  - **Alternative hypothesis (H_A):** Opposes the null hypothesis, indicating a real change, difference, or effect beyond random chance.
- The goal is to use sample data to determine whether there is sufficient statistical evidence to reject H_0 or fail to reject H_0 in favor of H_A.

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Previously… General strategy for hypothesis testing

1. **Calculate the observed test statistic:** Compute the summary metric (e.g., difference in means) from your actual sample data.
   - Identifying the appropriate statistic is critical for capturing the effect.
   - This is the **observed statistic** under the null hypothesis.
2. **Construct the null distribution:** Use permutations or bootstrapping to simulate the test statistic's behavior **assuming the null hypothesis is true**.
   - This creates a distribution of outcomes that could occur purely by chance.
3. **Evaluate plausibility (Calculate p-value)**: Compare the **observed statistic** to the **null distribution** to see how "rare" the result is.
   - **The p-value**: How likely are we to observe a value as extreme as (or more extreme than) the observed statistic if the null hypothesis were true?

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Example

- Suppose we have two samples that represent Biki ride durations for two distinct neighborhoods *a* and *b*.
- Instead of working with actual data, sample the data for the two neighborhoods from a normal distribution with a mean ($\mu = 20$) and a standard deviation ($\sigma = 4$).
  - We will sample 15 bike rides for each of the neighborhoods *a* and *b*.
  - In our scenario, this amounts to measuring the duration of 30 bike rides in neighborhoods *a* and *b* combined.

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Example (cont'd.)

1. Compute and compare the means of datasets *a* and *b*.
   - Based on the observed difference between the means, do you believe there is a statistically significant difference between samples *a* and *b*?

```
np.random.seed(0)
a = np.random.normal(20, 4, 15)
b = np.random.normal(20, 4, 15)
print(f"({np.mean(a):.4f}, {np.mean(b):.4f})")
print(f"Difference of means: {np.mean(a) - np.mean(b):.2f}")
```

```
(22.7480, 20.7948)
Difference of means: 1.95
```

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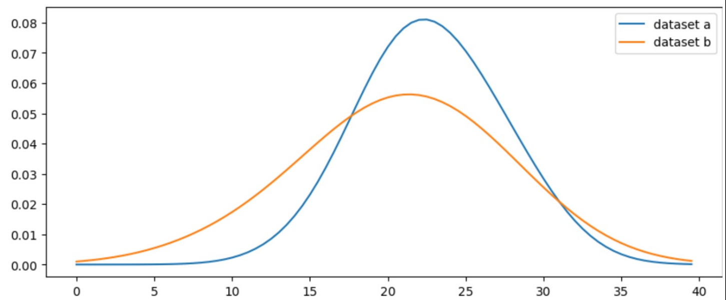
Example (cont'd.)

- The two KDEs representing the datasets *a* and *b*.

```python
import scipy.stats as stats

x_axis = np.arange(0, 40, 0.5)
kde_1 = sp.stats.gaussian_kde(a, bw_method=1)
kde_2 = sp.stats.gaussian_kde(b, bw_method=1)
x_densities_1 = kde_1.evaluate(x_axis)
x_densities_2 = kde_2.evaluate(x_axis)
```

```python
plt.figure(figsize=(10, 4))
plt.plot(x_axis, x_densities_1, label="dataset a")
plt.plot(x_axis, x_densities_2, label="dataset b")
plt.legend();
```



- Visually, does dataset *a* differ significantly from dataset *b*?

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Example (cont'd.)

2. Construct the null distribution using the resampling method.
   - Plot a histogram of the difference of means obtained during resampling and the observed value.
3. Evaluate plausibility by comparing the observed statistic to the null distribution to determine how "rare" it is.
   - Provide a p-value to support your claim.

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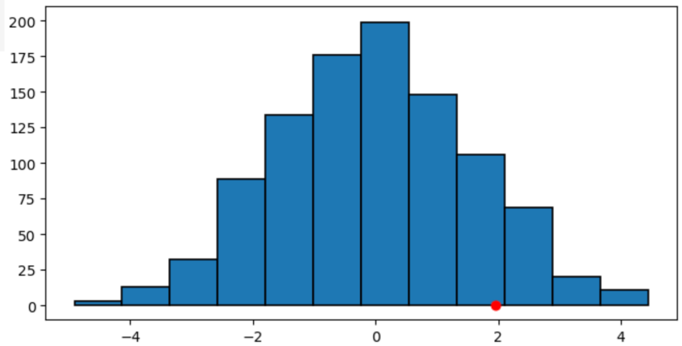
```
all_data = np.concatenate([a, b])
means_difference = []

for _ in range(1000):
    np.random.shuffle(all_data)
    a_subset = all_data[:15]
    b_subset = all_data[15:]
    means_difference.append(np.mean(a_subset) - np.mean(b_subset))

plt.figure(figsize=(8, 4))
plt.hist(means_difference, edgecolor='k', linewidth=1.2, bins=12)
plt.scatter(np.mean(a) - np.mean(b), 0, color="r", s=40)
plt.ylim(-10, 210)
plt.show()
```



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Example (cont'd.)

- Given the relatively large p-value, we fail to reject the null hypothesis, indicating the difference between the sample means is not statistically significant.

```
obs_mean_diff = np.mean(a) - np.mean(b)
sum(means_difference >= obs_mean_diff)
```

```
118
```

```
sum(means_difference >= obs_mean_diff) / len(means_difference)
```

```
0.118
```

```
both_sides = np.abs(np.array(means_difference)) >= np.abs(np.array(obs_mean_diff))
sum(both_sides) / len(means_difference)
```

```
0.228
```

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Comparing a sample to a distribution parameter

- Ex. The national average for the math score of the SAT is 525, with a standard deviation of 20. Does a sample of 100 students who followed your curriculum significantly differ from the national average?
  1. We can generate bootstrap samples and bootstrap statistics (replicates) and infer the 95% confidence interval for the national SAT scores.
  2. If the sample mean of the students who followed your curriculum is included in the 95% confidence interval, then we don't have enough evidence to discard the null hypothesis.
     - I.e., the scores obtained by our 100 students are from the theoretical distribution with $\mu = 525$ and $\sigma = 20$.

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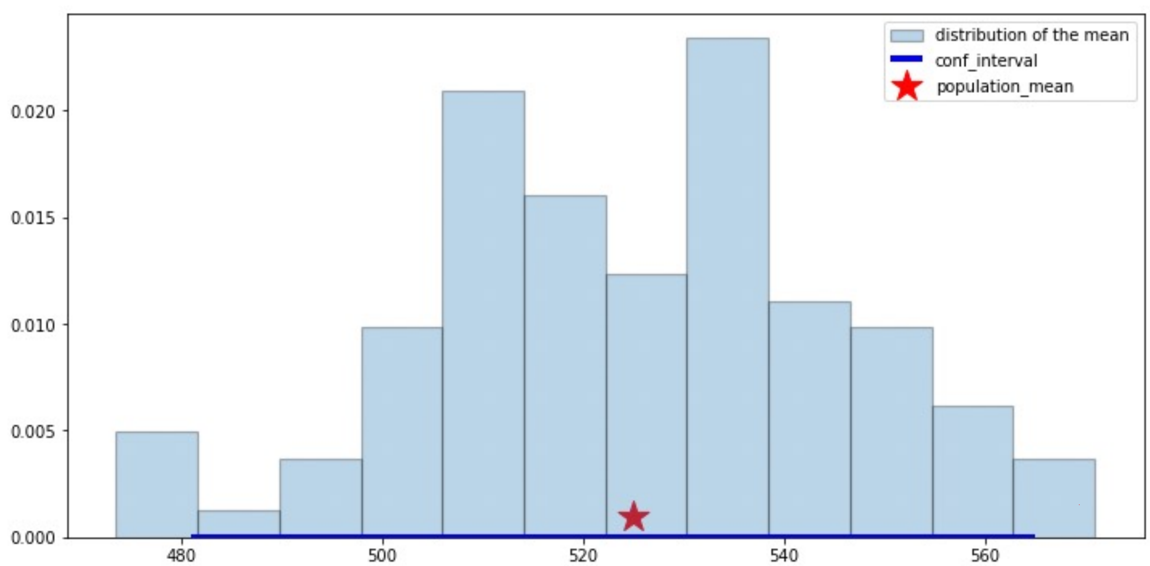

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Comparing proportions

- Assume that the national proportion of branded to unbranded generic medications is 1:5.
- An NGO is exploring incentives to increase use of unbranded generics as to control healthcare spending.
  - The incentives are tested in a geographical area for one year.
  - In a survey of the medication sold in that area, out of the 4,434 drugs sold, 3,766 were generics.
- Is the result different from the national proportion of branded to unbranded generic medications?
  - Note that we are merely asking whether the results are different, not whether the program worked.

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Testing proportion

- A ratio of 1:5 means that 5/6 or ~ 83% of drugs are generic.
- After implementing a strategy to increase the usage of generics, 3,766 out of 4,434 (84.9%) drugs were generic.
- Could this difference be due to random sampling alone or may the strategy be causing the result?
- We examine the distance between 83% (national ratio) and the observed sample proportion of 84.9%.
  - Larger distances provide stronger evidence that the strategy may be having an effect.

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Deciding on the test statistic

- We can use the absolute distance between the national proportion and that obtained from the survey.

$$|(\text{sample \% of generic drugs}) - 83|$$

- **Null hypothesis**: There is no difference between sample percentage of generic drugs sold and 83.
  - I.e., the observed difference is merely due to sampling variability.
- **Alternative hypothesis**: The difference is unlikely to have arisen due to chance alone.

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Building the null distribution

- What are the plausible values of the test statistic under the null hypothesis?
- If we were to sample data from outside of the testing area, a difference of the samples and national mean look like:
  - G: Generic, B: Brand Name

	Difference with National rate
Sample 1 `G, B, G, G, … B` % Generic = 84%	|84 – 83| = 1
Sample 2 `G, G, B, B, … G` % Generic = 85%	|8 5– 83| = 2
Sample 3 `B, B, G, G, … G` % Generic = 82%	|8 2– 83| = 1
…	…
Sample n `G, B, G, B, … G` % Generic = 83%	|8 3– 83| = 0

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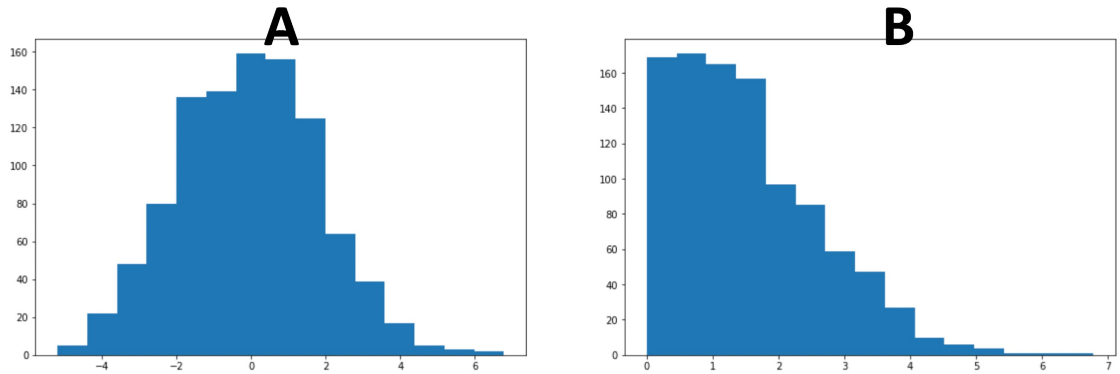
Question

- What would the resulting null distribution of the test statistic look like?



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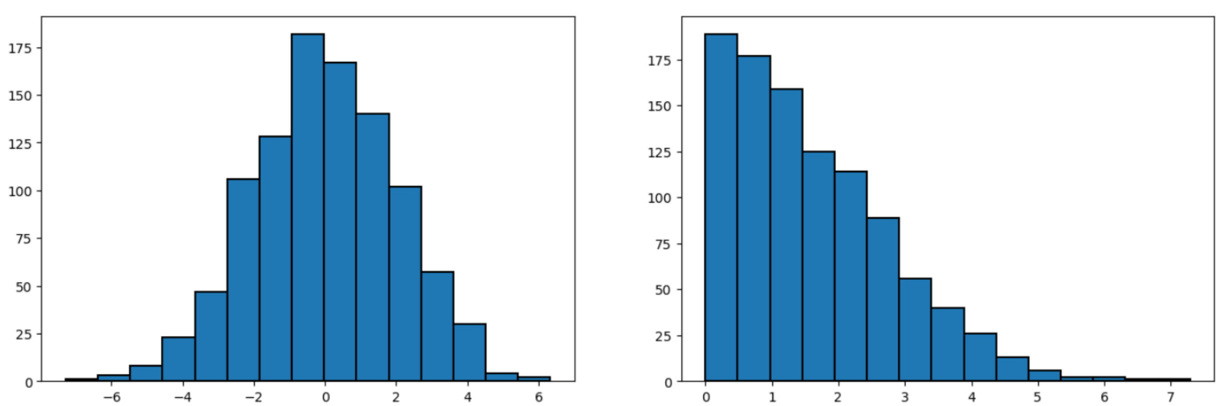
```python
data = np.random.normal(0, 2, 1000)

plt.figure(figsize=(16, 5))

plt.subplot(1, 2, 1)
plt.hist(data, bins=15, edgecolor='k', linewidth=1.4)
plt.subplot(1, 2, 2)
plt.hist(np.abs(data), bins=15, edgecolor='k', linewidth=1.4);
```



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Question

- Beyond collecting another dataset from the surveyed area, what other strategies can we use to explore the range of difference between 83% and samples that come from the same distribution?
  - I.e., how can we generate random samples that look like those produced under the null hypothesis without physically sampling new data?


```
# First example
model_proportions = [0.17, 0.83]

data = np.random.choice(["B","G"], p=model_proportions, size=4434)
data
```

```
array(['G', 'G', 'G', ..., 'G', 'G', 'G'], shape=(4434,), dtype='<U1')
```

```
print(sum(data == "G") / 4434 * 100)
```

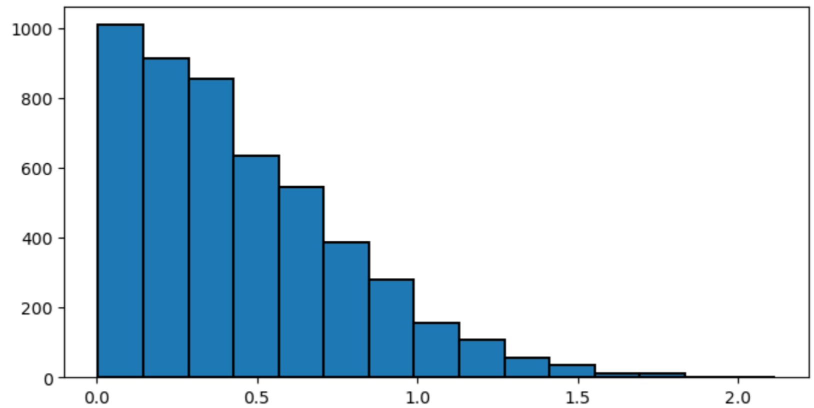
```
83.13035633739287
```

```
# Or using the binomial distribution
np.random.binomial(4434, 0.83) / 4434 * 100
```

```
82.83716734325665
```

```python
sample_diffs_null = []
for _ in range(5000):
    sample_proportion = sum(np.random.choice([0, 1], p=model_proportions, size=4434)) / 4434 * 100
    sample_diff = np.abs(sample_proportion - 83)
    sample_diffs_null.append(sample_diff)

plt.figure(figsize=(8, 4))
plt.hist(sample_diffs_null, bins=15, edgecolor='k', linewidth=1.4);
```



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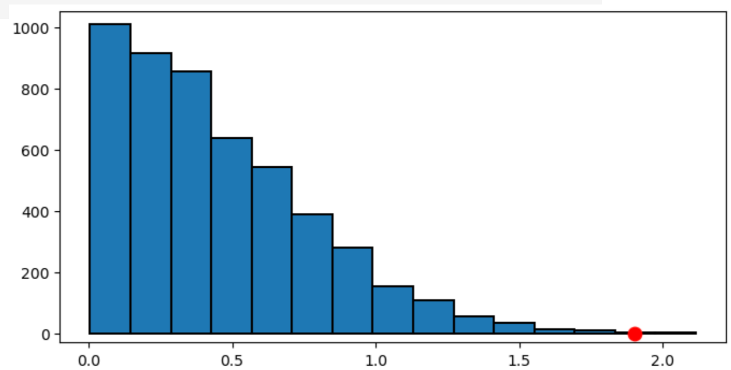
```python
plt.figure(figsize=(8, 4))

plt.hist(sample_diffs_null, bins=15, edgecolor='k', linewidth=1.4)
plt.scatter(1.9, 0, color='r', s=80)
plt.ylim(-30, 1050);
```




```python
# computimg the p-value
p_value = sum(np.array(sample_diffs_null) >= 1.9) / len(sample_diffs_null)
p_value
```

```
0.0002
```

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Comparing multiple categories

- The method we used in the previous example extends to models with multiple categories.
- The approach will be the same as before.
  - The only difference is that we have to come up with a new test statistic.

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Example: New method for estimating fish diversity?

- You have a faster, autonomous machine learning-based method of estimating the diversity of fish in a given area.
- Your method can provide ratios for Tilapia, Blenny, Angelfish, Salmon and Other in a pisciculture pond.
- You manually collected 500 and counted each ratio.
  - This tedious method was the state of the art at that time.
- You used the eDNA to estimate the ratios.
- Results are shown in the table and graph on the following two slides.

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```
fish_proportions = pd.DataFrame({
    'kind': ['Tilapia', 'Blenny', 'Angelfish', 'Salmon', 'Other'],
    'sampled': [0.20, 0.08, 0.12, 0.54, 0.06], 'eDNA': [0.26, 0.08, 0.08, 0.54, 0.04]})
fish_proportions
```

	kind	sampled	eDNA
0	Tilapia	0.20	0.26
1	Blenny	0.08	0.08
2	Angelfish	0.12	0.08
3	Salmon	0.54	0.54
4	Other	0.06	0.04

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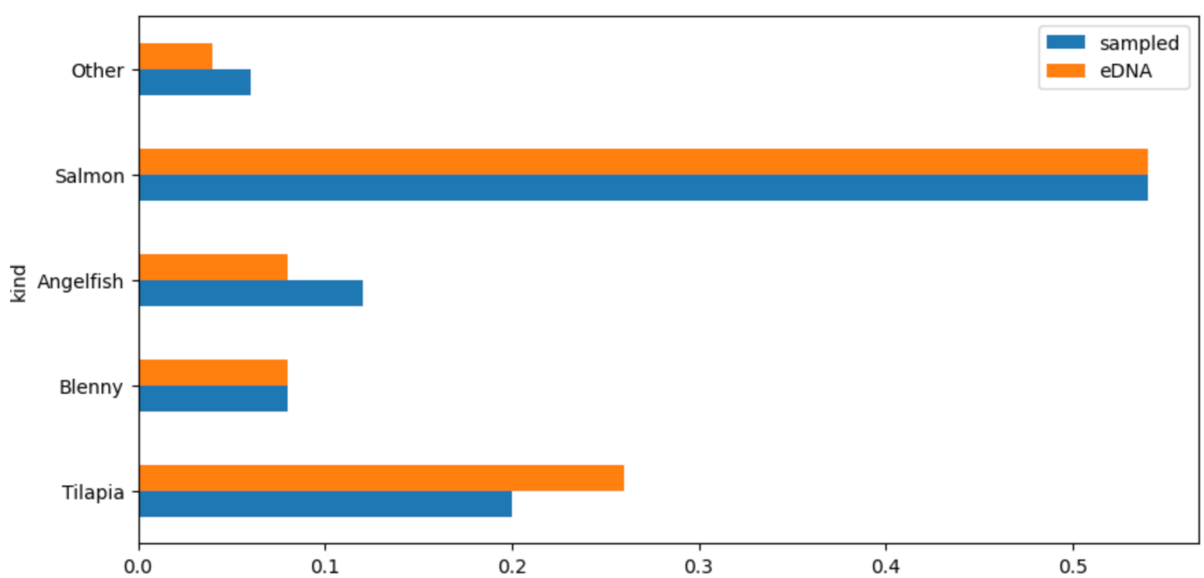
```
fish_proportions.plot.barh(x="kind", figsize=(10, 5));
```



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Questions

- The data distributions look slightly different.
- We need to answer the following questions:
  - What are the null and alternative hypotheses?
  - What is a useful test statistic for this problem?
  - How can we generate data following the null hypothesis?

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1. Null and alternative hypotheses

Our hypotheses are:

- Null hypothesis: The observed difference between both methods is due to sampling variability.
  - If we were to sample again, we may see similar or even greater difference due to random variation.
- Alternative hypothesis: The difference is too big to be due to random chance alone.
  - If we were to sample again, we would rarely, if ever, observe such a difference.
- Note that here again, we are not concerned in whether or not our method is over or underestimating a specific fish.
  - We only care about whether or not the method yields equivalent results to the counting a large sample manually.

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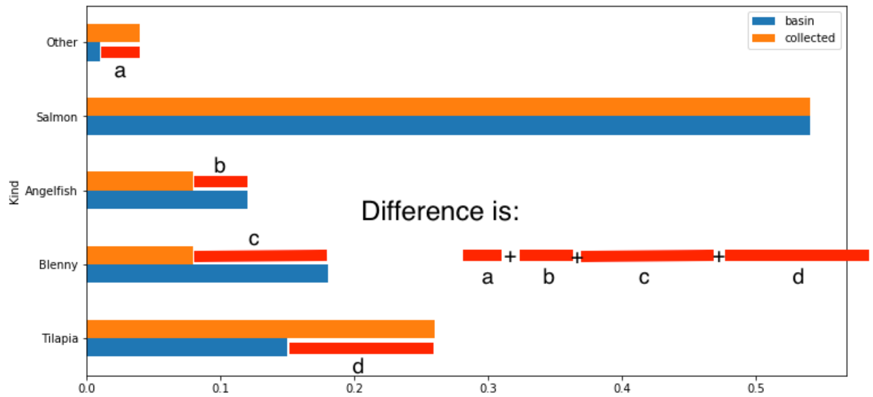
2. The test statistic

- A good test statistic for this problem is one that allows us to quantify the difference between the two distribution.



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The distance between two distributions

- We can quantify the distance between two distributions using the **total variation distance**.
  - Conceptually similar to the Euclidean distance between two vectors.

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```python
fish_proportions["difference"] = fish_proportions["sampled"] - fish_proportions["eDNA"]
fish_proportions
```

	kind	sampled	eDNA	difference
0	Tilapia	0.20	0.26	-0.06
1	Blenny	0.08	0.08	0.00
2	Angelfish	0.12	0.08	0.04
3	Salmon	0.54	0.54	0.00
4	Other	0.06	0.04	0.02

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The total variation distance (TVD)

- Notice that the sum of the entries of `difference` is 0.
  - The positive entries add up to 0.06, exactly canceling the total of the negative entries which is –0.06.
- `basin (sampled)` and `collected (eDNA)` add up to 1.
  - 1 – 1 needs to be equal to zero.
- To avoid numbers canceling each other, we sum the absolute values and divide by two:

$$tvd = \frac{1}{2} \sum_{x \in \Omega} |P(x) - Q(x)|$$

where P and Q are the two probability measures over Ω.

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```python
observed_tvd = 1/2 * sum(np.abs(fish_proportions['sampled'] - fish_proportions['eDNA']))
observed_tvd
```

```
0.06
```

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3. Generating data under the null hypothesis

- Given two categories, `Tilapia` and `Other`, and proportion of each, p(`Tilapia`) = 0.2 and p(`Other`) = 1 – p(`Tilapia`) = 0.8.
  - We can generate 200 instances (Tilapia or Other) using the binomial distribution.

```python
sample_1 = np.random.choice(['Tilapia', 'Other'], p=[0.20, 0.80], size=500)
print(sample_1[0:10])
print(sum(sample_1 == "Tilapia"))
```

```
['Other' 'Other' 'Other' 'Other' 'Tilapia' 'Other' 'Other' 'Other' 'Other'
 'Other']
102
```

```python
print(np.random.binomial(500, 0.20))
print(np.random.binomial(500, 0.20))
```

```
97
104
```

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The multinomial distribution

- The multinomial distribution generalizes the binomial distribution to multiple categories.
  - Each trial is independent of the others.
  - The probability of the outcomes remain constant across trials.
  - The experiment consists of *n* repeated trials.
  - Each trial results in one of several discrete possible outcomes (the categories).

```
# Throw a dice 120 times
np.random.multinomial(120, pvals=[1/6, 1/6, 1/6, 1/6, 1/6, 1/6] )
```

```
array([22, 20, 19, 19, 17, 23])
```

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```
# This is equivalent to the multinomial returning the outcomes of each trial
np.random.choice(['Tilapia', 'Blenny', 'Angelfish', 'Salmon', 'Other'],
                 p=[0.20, 0.08, 0.12, 0.54, 0.06], size=500)
```

```
array(['Salmon', 'Salmon', 'Salmon', 'Angelfish', 'Salmon', 'Salmon',
       'Salmon', 'Salmon', 'Tilapia', 'Salmon', 'Blenny', 'Salmon',
       'Salmon', 'Salmon', 'Salmon', 'Salmon', 'Salmon', 'Salmon',
       'Salmon', 'Other', 'Salmon', 'Salmon', 'Salmon', 'Tilapia',
       'Blenny', 'Blenny', 'Tilapia', 'Angelfish', 'Angelfish', 'Blenny',
       'Salmon', 'Salmon', 'Salmon', 'Angelfish', 'Tilapia', 'Salmon',
       'Salmon', 'Tilapia', 'Salmon', 'Salmon', 'Blenny', 'Salmon',

                                  ⋮

       'Tilapia', 'Salmon', 'Salmon', 'Salmon', 'Blenny', 'Salmon',
       'Salmon', 'Other', 'Tilapia', 'Other', 'Salmon', 'Salmon',
       'Salmon', 'Tilapia', 'Salmon', 'Salmon', 'Salmon', 'Blenny',
       'Blenny', 'Salmon', 'Blenny', 'Blenny', 'Tilapia', 'Tilapia',
       'Salmon', 'Tilapia', 'Salmon', 'Other', 'Salmon', 'Tilapia',
       'Tilapia', 'Angelfish', 'Tilapia', 'Salmon'], dtype='<U9')
```

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4. Simulating the experiment

- Given the sample proportions, what is the sampling distribution of the tvd?
    1. Generate a random sample using the multinomial distributions and the observed proportions from manual method.
    2. Compute and store the sample tvd.

```
n = 500
samples_tvd = []

for i in range(5000):
    sample_proportions = np.random.multinomial(n, pvals=fish_proportions["sampled"]) / n
    sample_tvd = 1/2 * np.sum(np.abs(fish_proportions['sampled'] - sample_proportions))
    samples_tvd.append(sample_tvd)
```

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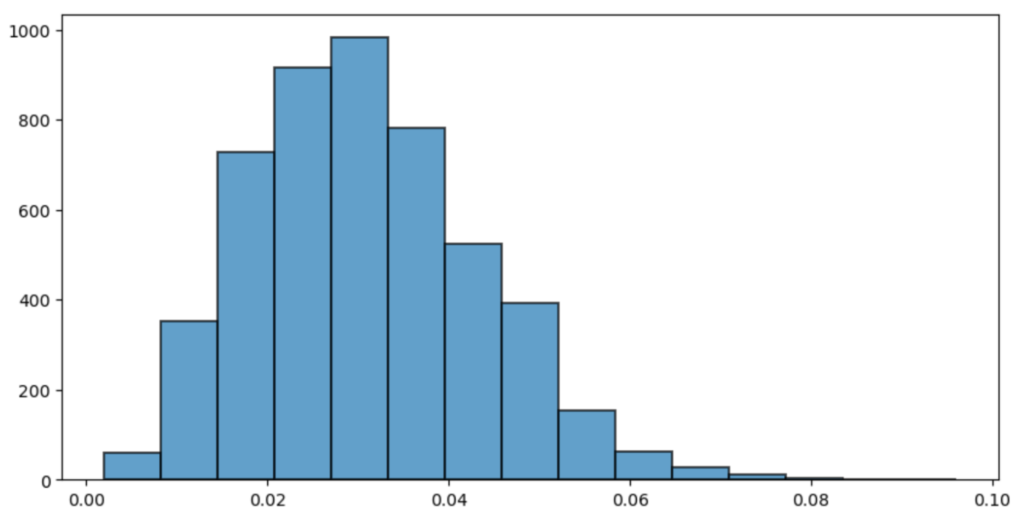
```
plt.figure(figsize=(10, 5))
plt.hist(samples_tvd, bins=15, edgecolor='k', linewidth=1.4, alpha=0.7);
```



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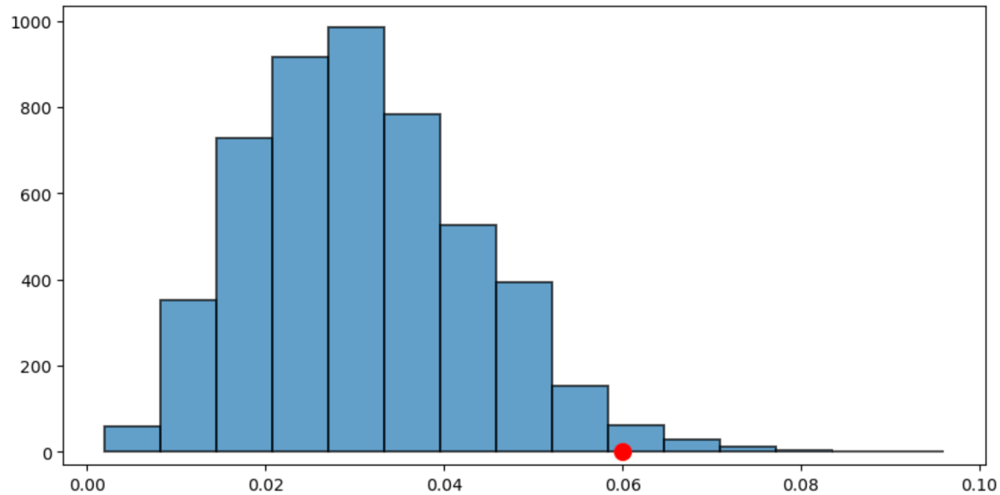
```
plt.figure(figsize=(10, 5))

counts = plt.hist(samples_tvd, bins=15, edgecolor='k', linewidth=1.4, alpha=0.7)
plt.scatter(observed_tvd, 2, color='r', s=100)
plt.ylim(-30, counts[0].max() + 50);
```



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Notes about the outcome

- Using a p-value threshold (significance level) of 0.05 the distribution of fish kind is different using both methods.
  - `tvd` of 0.06 is far out in the tail of the histogram.

```
p_value = np.sum([np.array(samples_tvd) >= observed_tvd]) / len(samples_tvd)
print(p_value)
```

```
0.0198
```

- The difference does not say *why* the distributions are different or what the difference might imply.

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